Abnormal Within-Spread Trading Volume and Short-Horizon Price Dynamics

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1. Introduction

Short-horizon price dynamics are difficult to interpret because transaction prices are formed inside a market with explicit trading frictions. In a frictionless asset-pricing view, a price change mainly reflects new information about fundamentals. In an actual trading venue, however, the transaction price also reflects immediacy demand, inventory absorption, adverse selection, and the willingness of liquidity suppliers to stand on the other side of order flow. The bid–ask spread is therefore not merely a mechanical difference between two quotes. It is an economic object that records the cost of immediate execution and the compensation required by liquidity suppliers for providing marketability. Dealer and adverse-selection models make this point directly: spreads compensate market makers for inventory and order-processing costs, while asymmetric information requires quotes to adjust to the possibility that incoming trades are informed [35, 18, 26].

The same observed amount of trading can arise from liquidity traders, institutional liquidation, or temporary demand shocks. These different sources of volume need not have the same implication for future returns. Some trades may reveal information and therefore support continuation, while others may reflect risk-sharing or temporary liquidity pressure and therefore support reversal. Theoretical work on volume emphasizes that trading activity can contain speculative information or risk-sharing motives [7, 28]. For this reason, treating volume as a single aggregate variable can obscure the economic channel through which trading affects prices.

The ambiguity becomes even sharper at high frequency. At the transaction level, the relevant object is not only how much volume occurs, but where that volume occurs relative to the prevailing bid and ask. A trade above the quoted midpoint is closer to buy-side immediacy demand because the transaction occurs nearer to the ask. A trade below the midpoint is closer to sell-side immediacy demand because the transaction occurs nearer to the bid. This distinction is consistent with the broader trade-classification and price-impact literature, which shows that matching trades to prevailing quotes is necessary for recovering the direction and informational content of trading activity [27, 21, 6]. If transaction location contains information about the side and urgency of trading demand, then above-midpoint and below-midpoint volume should not automatically be pooled into a single total-volume measure.

This paper is motivated by the gap between aggregate-volume studies and order-flow studies. Aggregate-volume studies document that volume is related to returns and volatility, but they often do not separate the different economic sources of volume. Order-flow studies show that buy–sell imbalance affects liquidity and returns, but they often focus on imbalance itself rather than asking whether the meaning of imbalance depends on the price state in which it appears. Yet the same abnormal sell-side volume may have very different implications in a market that is already drifting downward than in a market whose pre-event price path is primarily noise-dominant. In the first case, liquidity suppliers absorb sell pressure while prices already contain a directional component. In the second case, abnormal selling arrives against a non-directional baseline. Evidence that order imbalance affects liquidity and market returns even after controlling for aggregate volume suggests that this side-specific decomposition is economically meaningful [9].

The central idea of this paper is therefore that volume becomes more informative and deviates the price path when it is conditioned along two dimensions. The first dimension is the baseline price state. Using a diffusion-based hypothesis-testing framework, the paper classifies pre-signal price sequences as drift-dominant or noise-dominant. The second dimension is the execution location of abnormal volume. Conditional on the detected baseline state, the paper asks whether abnormal consolidated volume appears above, at, or below the bid–ask midpoint. The empirical object is not simply whether high volume predicts future returns, but whether abnormal within-spread volume causes the future price path to follow or deviate from the path implied by the baseline market state.

This design is motivated by three connected strands of literature. From market microstructure, it takes the view that transaction prices and spreads reveal immediacy costs, inventory risk, and adverse-selection concerns [14, 30, 24]. From the volume literature, it takes the view that trading activity is informative but ambiguous unless the source of trading is identified [1, 25, 15]. From the order-flow literature, it takes the view that signed trading pressure matters beyond aggregate volume and that liquidity has systematic common components [10, 9]. The paper combines these insights by studying abnormal volume inside the spread, conditional on whether the preceding price path is drifting or noise-dominant.

2. Related Work

The central problem in this paper is that trading volume is informative but not self-interpreting. A given amount of volume may reflect information arrival, liquidity demand, inventory transfer, institutional rebalancing, or behavioral liquidation. These mechanisms can generate similar trading intensity but different future price paths. For this reason, aggregate volume is often too coarse to identify the economic force behind short-horizon return dynamics. The paper therefore treats volume as informative only after two additional pieces of structure are introduced: the baseline state of the price process and the location of execution within the bid–ask spread.

The first piece of structure comes from market microstructure. Transaction prices are not formed in a frictionless environment; they are shaped by immediacy costs, inventory risk, and adverse selection. The bid–ask spread compensates liquidity suppliers for standing ready to trade when incoming orders may impose inventory risk or contain private information. Once the spread is understood in this way, the location of a trade inside the spread becomes economically meaningful. A transaction near the ask is closer to buy-side immediacy demand, while a transaction near the bid is closer to sell-side immediacy demand. This motivates the paper's use of above-midpoint, at-midpoint, and below-midpoint volume rather than total volume alone [35, 18, 26].

This logic also explains why high-frequency trade and quote alignment matters. The side and urgency of a transaction cannot be inferred from volume alone; the trade must be interpreted relative to the prevailing quote. If trades are not matched to the relevant bid and ask, execution-location measures may confuse genuine buying or selling pressure with quote-timing noise. The trade-classification and execution-cost literature therefore provides the empirical foundation for the paper's data construction: consolidated trades are aligned with prevailing quotes and then classified by whether they occur above, at, or below the midpoint [27, 6].

A second piece of structure comes from the fact that trades can affect prices through more than one channel. Transaction-level trading may reveal private information, impose inventory pressure, or interact with public information shocks. As a result, abnormal volume should not be interpreted independently of the price environment in which it appears. The same below-midpoint abnormal volume may have a different meaning if the pre-signal price path already contains detectable drift than if the price path is noise-dominant. In a drift-dominant state, abnormal sell-side execution may reinforce an existing directional pressure; in a noise-dominant state, it may represent only a temporary disturbance. This motivates the paper's decision to identify the baseline price state before evaluating the abnormal-volume signal [14, 30, 21, 24, 29].

Moreover, volume can reveal information beyond prices, but it can also reflect risk-sharing trades or temporary liquidity pressure. This is exactly why the empirical volume–return relation can appear unstable across settings. If speculative volume and risk-sharing volume have different implications, then the proper empirical object is not aggregate volume itself, but volume classified by the context in which it occurs [1, 15, 25].

Order imbalance and price-pressure studies provide the closest bridge from aggregate volume to the paper's within-spread measure. Directional pressure matters because buy- and sell-side trading are not symmetric once liquidity suppliers must absorb order flow. Total volume measures trading intensity, but imbalance measures the direction of pressure. The present paper uses execution location relative to the midpoint as a practical way to recover this side-specific information from consolidated trades. Above-midpoint abnormal volume is treated as closer to buy-side pressure; below-midpoint abnormal volume is treated as closer to sell-side pressure. This design follows the broader evidence that the structure and direction of trading activity carry information beyond total volume [10, 9, 4].

Behavioral finance adds another reason to expect the effect of abnormal volume to depend on the recent price path. If investors evaluate positions relative to reference prices, then trading after gains, losses, or persistent directional movement need not be symmetric. Realization decisions, delayed selling, underreaction, and reference-dependent adjustment can all affect when trading pressure appears and how long its price effect lasts. This behavioral channel does not replace the liquidity-pressure mechanism; rather, it helps explain why abnormal within-spread volume may have different predictive content after a drifting sequence than after a noise-dominant sequence. It also gives a reason why below-midpoint pressure may be especially informative when the recent path has already created losses, liquidation pressure, or reference-point effects [32, 31, 16, 19].

The paper therefore sits at the intersection of these literatures. Microstructure theory explains why execution location is meaningful. Trade-and-quote work explains how that location can be measured. Volume models explain why aggregate volume is ambiguous. Order-flow studies explain why side-specific pressure matters. Behavioral finance explains why the same pressure may be stronger in some price states than in others. The empirical contribution is to combine these ideas in one conditioning framework: first classify the baseline price state, then measure abnormal volume by location inside the spread, and finally test whether the future price path follows or deviates from that baseline.

3. Conflicted result on short term return

Campbell, Grossman, and Wang show that the first daily autocorrelation of stock returns is lower on high-volume days than on low-volume days. Their interpretation is that unusually high volume is more likely to reflect noninformational trading pressure that is temporarily absorbed by risk-averse market makers, so price changes on those days are more likely to be followed by reversal than price changes on low-volume days. They argue this pattern is not just an artifact of nonsynchronous trading, because it also appears in large indexes and individual large stocks [8].

Gervais, Kaniel, and Mingelgrin [17] conclude that unusual trading volume predicts the direction of future stock returns: high-volume stocks subsequently outperform and low-volume stocks underperform, suggesting that volume contains information about future price evolution beyond that in past returns.

The two papers are close in motivation, but they are different in statistical design. The conflicting empirical findings suggest that the volume premium literature still contains an important unresolved gap. If there were a single homogeneous underlying relation between volume and future price movement, differences in methodology could affect magnitude, noise, and statistical significance, but they should not repeatedly reverse the sign of the estimated relation. In particular, if the underlying relation were genuinely positive, then a high-volume condition should not systematically produce a decline in the autoregressive dependence of returns; at a minimum, it should not generate a reversal-type pattern so persistently. Conversely, if the reversal-type relation fully characterized the true effect of volume, then a positive volume premium should not arise so regularly. This suggests that the observed volume variable is likely mixing economically different market regimes rather than capturing one common effect.

Several features of the existing literature may contribute to this instability. First, different assets can exhibit very different trading behavior, liquidity structure, volatility, and responses to order flow, so the same observed volume pattern need not carry the same implication across stocks. Second, existing designs typically do not condition on the underlying market state, even though the economic meaning of a given volume realization may differ sharply depending on whether the price process is upward drifting, downward drifting, or primarily noise dominated. Third, they usually do not distinguish where volume is executed relative to the bid-ask spread, even though volume near the ask, near the bid, and near the midpoint may reflect very different forms of trading pressure. When these heterogeneous situations are pooled together, the estimated relation between volume and future price movement can become unstable and may appear with different signs across studies.

These gaps suggest that further investigation of the volume premium should not rely only on alternative aggregate definitions of volume, but should instead separate the contexts in which volume is observed. The same amount of consolidated trading volume may have different implications when the market is in an upward-drifting state, a downward-drifting state, or a noise-dominated state. Moreover, the location of that volume relative to the bid-ask spread should also matter. Volume executed near the ask may indicate stronger willingness to pay for immediacy on the buy side, whereas volume executed near the bid may indicate stronger willingness to accept immediacy cost on the sell side. If these different forms of volume are pooled together, the resulting estimate can help explain why previous studies reach conflicting conclusions and provide a framework on how to study this kind of problem more detailed and concretely.

4. Data and methods

4.1 data source and processing

Our empirical analysis uses raw high-frequency consolidated trade and quote data from the WRDS TAQ database. At the event level, each consolidated trade is aligned with the quote at the real-time millisecond timestamp, so that the prevailing bid–ask information is tracked as accurately as possible at the moment of execution. The event-level data are then aggregated to the one-minute level, to fit in the diffusion model and to produce the statistical measures we mentioned. Therefore, the resulting dataset counts how many trade volume happens at above/at/below midspread, and the prevailing vwap price.

In market microstructure, the bid is the highest displayed price at which market participants are currently willing to buy, while the ask is the lowest displayed price at which they are willing to sell. The bid–ask spread,

$$S_t = A_t - B_t,$$

Economically, buying at the ask means the trader is demanding immediacy on the buy side. They are willing to pay the higher posted selling price in order to obtain execution now rather than wait. Symmetrically, selling at the bid means the trader is demanding immediacy on the sell side, accepting the lower posted buying price to exit quickly. In that sense, trading at the ask or bid is not just a mechanical detail of execution; it reveals a willingness to pay a liquidity cost in exchange for speed. That is exactly why trade location relative to the spread can be informative about trading urgency and one-sided pressure.

This also explains why the spread matters. A wider spread implies a larger cost of immediate execution and therefore lower liquidity. The classic result treats the spread as an economically meaningful transaction cost and shows that assets with larger spreads tend to require higher expected returns, precisely because investors must be compensated for lower liquidity. More generally, microstructure research treats the spread as reflecting compensation for supplying immediacy under inventory risk, adverse selection, and other trading frictions.[22][11]

We tested on most actively traded large market cap stocks over 2020-2026, since this is an early draft and some data is still under processing, we test on only stock:

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"META",
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4.2 measures and modeling assumptions

Which is one of the magnificent 7 technology company. For a given minute, let P^A and P^B denote the prevailing ask and bid prices, respectively.

Quoted Bid–Ask Spread The observed quoted bid–ask spread S is defined as

$$S = P^A - P^B, \tag{1}$$

where P^A is the ask price and P^B is the bid price.

Quoted Midpoint A simple proxy for the center of the quoted market is the midpoint P^M, defined as

$$P^M = \frac{P^A + P^B}{2}, \tag{2}$$

where P^A and P^B are the raw ask and bid prices. Accordingly, the midpoint is also defined in raw form. We use the discrete midpoint value to split consolidated trades into three groups: above-mid trades, at-mid trades, and below-mid trades. A trade above the midpoint indicates that market participants are willing to bear some disadvantages to acquire the asset urgently. By contrast, a trade below the midpoint indicates some urgency to obtain liquidity. Trades executed at the midpoint can be interpreted as transactions taking place at a more balanced location within the spread.

Our conjecture is that imbalances across these three types of trading behavior contain information about underlying market conditions and may possess predictive power for subsequent price dynamics, which we examine in the following sections.

Trade Counts by Location Let P denote the observed transaction price. For each minute t, let N_t^+, N_t^0, and N_t^- denote the volume of consolidated trades above the midpoint, at the midpoint, and below the midpoint, respectively. These counts are defined as

$$\begin{cases} N_t^+ = \sum_j \mathbf{1}\{P_t^M < P_{t,j} < P_t^A\}, \\ N_t^0 = \sum_j \mathbf{1}\{P_{t,j} = P_t^M\}, \\ N_t^- = \sum_j \mathbf{1}\{P_t^B < P_{t,j} < P_t^M\}, \end{cases} \tag{3}$$

where $P_{t,j}$ denotes the price of trade j in minute t.

Quantile-Based Definition of Abnormal Activity Let N_t^k denote the count measure for trade-location category $k \in \{+, 0, -\}$ at time t. We adopt a quantile-based approach to define abnormal trading activity. Specifically, let Q^k denote an upper quantile of the empirical distribution of N_t^k over a historical reference sample. We classify the trading activity in category k as abnormal whenever

$$N_t^k > Q^k. \tag{4}$$

Thus, abnormality is defined relative to the historical distribution of the corresponding count measure. We adopt a quantile-based approach to define abnormal activity because trading activity is not directly comparable across categories, stocks, or time periods. It is also very worth to mention that the count distribution is roughly centered in a regular bulk but has a heavy right tail, then a quantile rule is well suited for identifying upper-tail extremes.

4.3 Price change

Future cumulative return We also define a cumulative return over a forward horizon of length H. Let P_t denote the price at the end of the detected segment and let P_{t+h} denote the price h periods ahead. The future cumulative return is

$$CR_{t,H} = \frac{P_{t+H} - P_t}{P_t} \tag{5}$$

This quantity summarizes the future price-change path over the post-event window and serves as the outcome variable in the subsequent empirical analysis.

4.4 Drift screening under a diffusion framework

We model the price process by

$$p_t = \int_0^t \mu_s \, ds + \int_0^t \sigma_s \, dW_s, \tag{6}$$

where μ_t is the local drift and σ_t is the instantaneous volatility.

In practice, drift is difficult to identify from observed returns because the deterministic component is often dominated by random variation. For the increment

$$r_{n,i} = p_{t_i} - p_{t_{i-1}}, \tag{7}$$

the relevant signal-to-noise ratio is

$$\frac{|\mu_t|\Delta_n}{\sigma_t\sqrt{\Delta_n}} = \frac{|\mu_t|\sqrt{\Delta_n}}{\sigma_t} \tag{8}$$

When this ratio is small, observed price movements do not clearly reveal whether the underlying mechanism is genuine drift or diffusion noise.

To address this problem, we follow the realized squared drift framework in [33]. The target quantity is the integrated squared drift over a detection window of length T_n,

$$ISD_{T_n} = \int_0^{T_n} \mu_s^2 \, ds. \tag{9}$$

The hypotheses are

$$H_0 : \ \mu_t \equiv 0 \text{ on } [0, T_n] \qquad \Longleftrightarrow \qquad H_0 : \ ISD_{T_n} = 0, \tag{10}$$

against

$$H_1 : \ \mu_t \not\equiv 0 \text{ on } [0, T_n] \qquad \Longleftrightarrow \qquad H_1 : \ ISD_{T_n} > 0 \tag{11}$$

The paper proposes the statistic

$$RSD_{T_n}(k) = \frac{1}{\Delta_n} \sum_{i=k+1}^{N_n} r_{n,i} r_{n,i-k}, \qquad N_n = \frac{T_n}{\Delta_n}, \tag{12}$$

which is the normalized k-lag return autocovariance. Its asymptotic variance is governed by the integrated quarticity

$$IQ_{T_n} = \int_0^{T_n} \sigma_s^4 \, ds \tag{13}$$

After standardization by a quarticity estimator, the resulting test statistic is asymptotically standard normal under the null and diverges under the alternative when

$$T_n \to \infty, \qquad \Delta_n \to 0, \qquad N_n = T_n/\Delta_n \to \infty \tag{14}$$

We assume that the fixed sample size detection window T is sufficiently long for the standardized statistic to provide an informative screen for nonzero drift magnitude. We also assume that the latent market process satisfies Assumptions 1–3 in [33], so that the diffusion structure and the associated asymptotic results are valid in our setting.

Abnormal-volume classification After market state has been identified, we classify daily trading activity using the location of consolidated trades relative to the prevailing midpoint of the bid–ask spread at the millisecond level. Each trade is labeled as above the midpoint, at the midpoint, or below the midpoint, and these location-specific trade volumes are aggregated to the daily level.

For each asset and each location-specific daily volume measure, abnormality is defined relative to the previous 60 trading days. Let V_d^+, V_d^0, and V_d^- denote daily trade volume above, at, and below the midpoint, respectively. For each measure, we compute a rolling upper-tail threshold based on the previous 60 trading days. In the benchmark specification, the cutoff is the rolling 90% quantile. A location-specific volume measure is classified as abnormal if it exceeds its own rolling threshold.

For each asset and each location-specific daily measure V_d^k, where $k \in \{+, 0, -\}$, abnormality is defined relative to its own recent historical distribution. Specifically, for each day d, we compute a rolling upper-tail threshold using the previous 60 trading days of the same asset and the same trade-location category. Let $Q_{d-1}^k(0.90)$ denote the rolling 90% quantile of V_τ^k over the preceding 60 trading days $\tau = d-60, \dots, d-1$. We then classify location-specific daily trading activity as abnormal whenever

$$V_d^k > Q_{d-1}^k(0.90).$$

5. Data summary

The dataset is constructed at the one-minute frequency and then grouped to the daily level for the descriptive summaries and abnormal-volume classification. For each minute, the data record the timestamp, the one-minute volume weight average price(VWAP), the prevailing bid and ask quotes, the bid–ask spread, and consolidated trade volume decomposed by execution location relative to the quoted midpoint: above the midpoint, at the midpoint, and below the midpoint.

table 1 reports daily summary statistics for the constructed META sample. The volume variables have 1,042 daily observations, while the daily VWAP percentage-change variable has 1,041 observations because one observation is removed due to stock split/combine. Total daily volume has a mean of about 15.67 million shares, a standard deviation of about 14.54 million shares, and a maximum above 212 million shares. This indicates that daily trading activity is highly dispersed rather than stable around a single average level.

Table 1: stock: "META" Daily Summary Statistics

Variable	n	Mean	SD	Min	Max
Volume above midpoint (thousands/day)	894	8,623.35	6,457.94	2,253.85	90,779.74
Volume below midpoint (thousands/day)	894	8,516.79	6,696.14	2,022.56	97,096.23
Volume at midpoint (thousands/day)	894	890.94	1,502.05	23.70	24,788.15
Price change (%/day)	893	0.1704	2.5050	-24.2223	25.3846
Total volume (thousands/day)	894	18,031.08	14,375.66	4,561.83	212,664.11

Note: Volume variables are reported in thousands of shares. Price change is daily VWAP percent change and is reported in percent.

The decomposition of daily volume shows a strong symmetry between the two directional sides of the spread. Above-midpoint volume has a mean of about 7.50 million shares, and below-midpoint volume has a mean of about 7.37 million shares. Their standard deviations are also close, about 6.60 million and 6.82 million shares, respectively. This similarity suggests that the constructed above/below midpoint measures are not mechanically dominated by only one side of the market. By contrast, midpoint or neither-side volume is much smaller, with a mean of about 0.80 million shares and a maximum of about 24.79 million shares. Therefore, most daily consolidated volume is classified away from the exact midpoint, which makes the above/below midpoint decomposition empirically meaningful for studying side-specific immediacy demand. The daily VWAP percentage-change variable is much more unstable than its mean alone suggests: its mean is positive, but the standard deviation is large and the maximum is extremely high relative to the rest of the table. For this reason, later empirical analysis should not rely only on unconditional price-change averages; it should condition on the baseline price state and compare future paths across abnormal-volume groups.

Figure 1 reports the percentage of daily observations with and without abnormal volume. The left bar is the no-abnormal-volume group, and the right bar is the abnormal-volume group. The figure shows that most trading days do not trigger the abnormal-volume screen: about 80.1% of days are classified as no-abnormal-volume days. The remaining 19.9% of days contain at least one abnormal-volume signal.

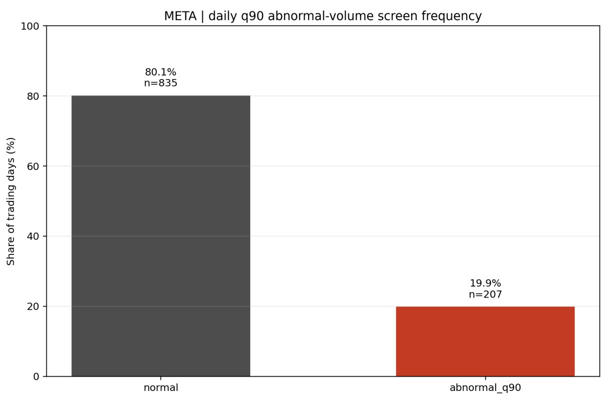

Figure 1: Share of daily observations of stock "META" with and without abnormal volume with either above midpoint or below midpoint

This split shows that trading volume is not stable over time. The abnormal-volume indicator is defined using a rolling 60-trading-day upper-tail screen, with the abnormal threshold set at the 90% quantile of the recent historical distribution. Even under this rolling benchmark, about one-fifth of the daily observations are still classified as abnormal. Unusually high trading activity is not a rare isolated event in the sample, rather, volume shifts frequently.

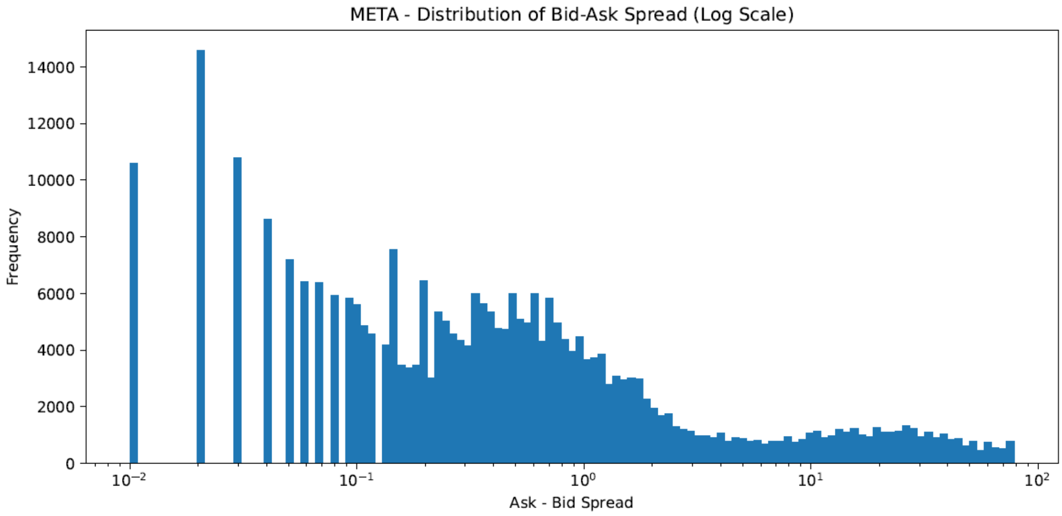

Figure 2: Distributional pattern of the quoted bid–ask spread.

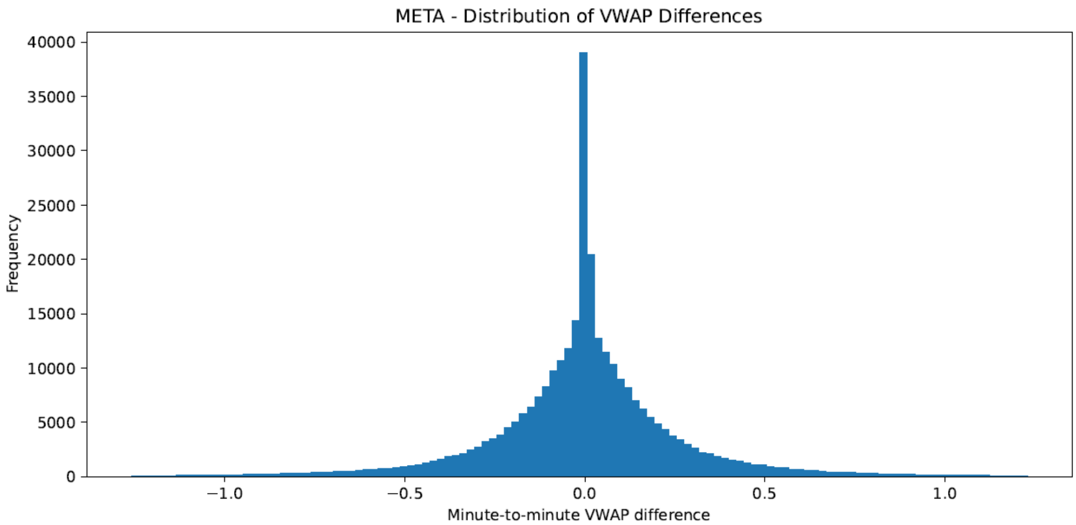

Figure 3: VWAP-based price variation used in the empirical construction.

Figure 2 reports the empirical distribution of the quoted bid–ask spread. The distribution is highly right-skewed. Most minute-level observations are concentrated very close to zero, while a small number of observations extend far into the right tail, with spreads reaching much larger values. This means that the typical liquidity environment is narrow-spread and highly liquid, but the sample also contains occasional high-friction episodes. The average spread alone would hide this asymmetry, because it would mix a large mass of normal low-spread minutes with a small number of unusually wide-spread minutes.

This shape is crucial for the paper's empirical design. In normal periods, active competition among liquidity suppliers keeps quoted spreads narrow. During high-friction periods, however, spreads can widen sharply as liquidity suppliers require compensation for order-processing costs, inventory exposure, and adverse-selection risk [24]. The long right tail therefore suggests that liquidity conditions are time-varying. Since quoted spreads contain market-wide time-series variation [10], the predictive content of above-midpoint or below-midpoint volume should be interpreted relative to the changing liquidity environment.

Figure 3 reports the distribution of minute-to-minute VWAP differences. The figure has a sharp peak around zero, meaning that most one-minute VWAP changes are very small. At the same time, the distribution has thick positive and negative tails, so large minute-level movements occur on both sides. This is exactly the empirical setting in which distinguishing drift from noise matters.

The thick tails of distribution imply that a subset of sequences may contain detectable price pressure. This motivates the later screening step: before interpreting abnormal within-spread volume, the paper first classifies whether the pre-signal price sequence is noise-dominant or drift-dominant.

Figures 4, 5, and 6 summarize the distribution of consolidated trade volume by execution location relative to the quoted midpoint. The x-axes use log-transformed volume, so zero-volume and very small-volume observations appear separately from the main active-trading mass. This representation is important because the raw volume variables are heavy-tailed and would otherwise be difficult to compare visually.

Figure 4 shows above-midpoint volume. The distribution has a visible mass near zero, followed by a sparse region of small positive values, and then a large concentrated hump at high log-volume values. This means that above-midpoint trading is not continuously distributed across all activity levels. Instead, many minutes have little or no above-midpoint activity, while active minutes cluster strongly around a high-volume region. Economically, this pattern is consistent with buy-side immediacy demand arriving in concentrated bursts rather than as a smooth background process.

Figure 6 shows a very similar pattern for below-midpoint volume. It also has a mass near zero, a thin middle region, and a large high-volume hump. The similarity between the above-midpoint and below-midpoint distributions is important. It suggests that the construction is not producing an artificial imbalance between buy-side and sell-side locations. Both sides appear to be generated by a similar liquidity-acquisition process, with many inactive minutes and a concentrated region of active trading.

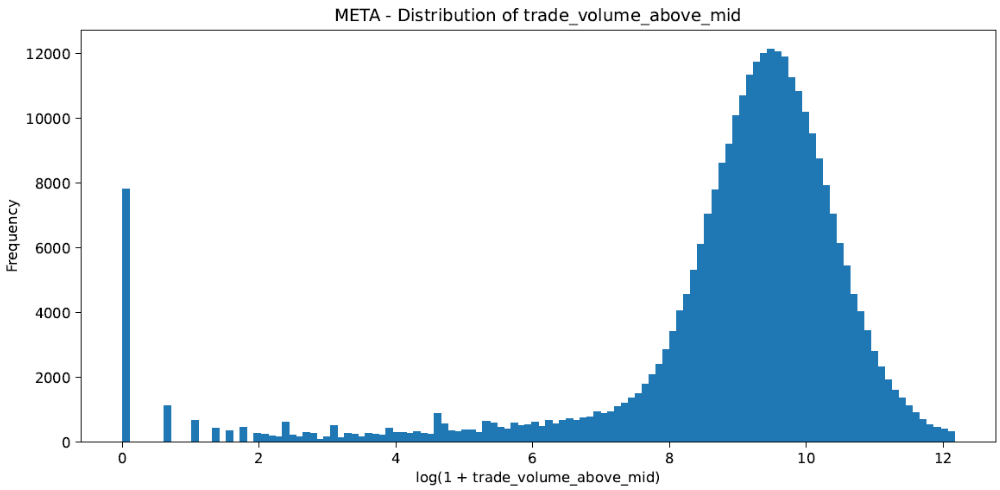

Figure 4: Trading volume above the quoted midpoint.

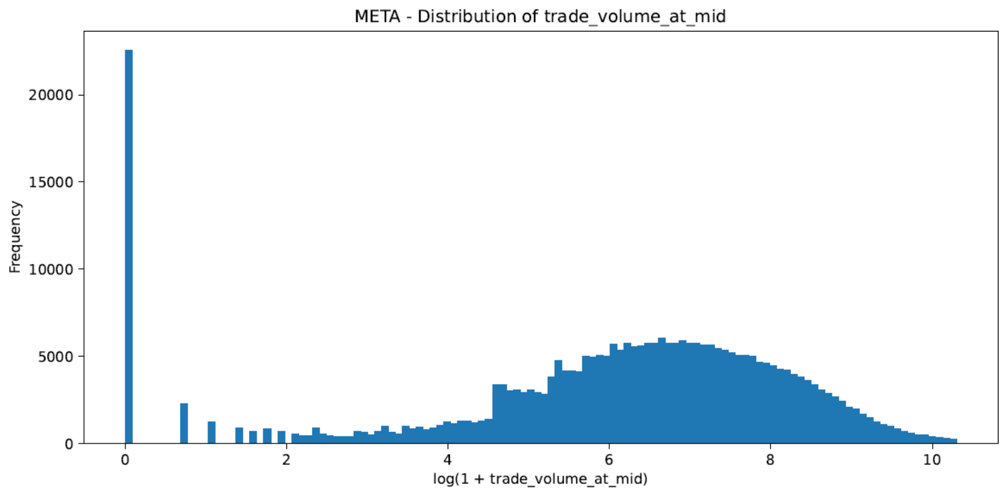

Figure 5: Trading volume executed at the quoted midpoint.

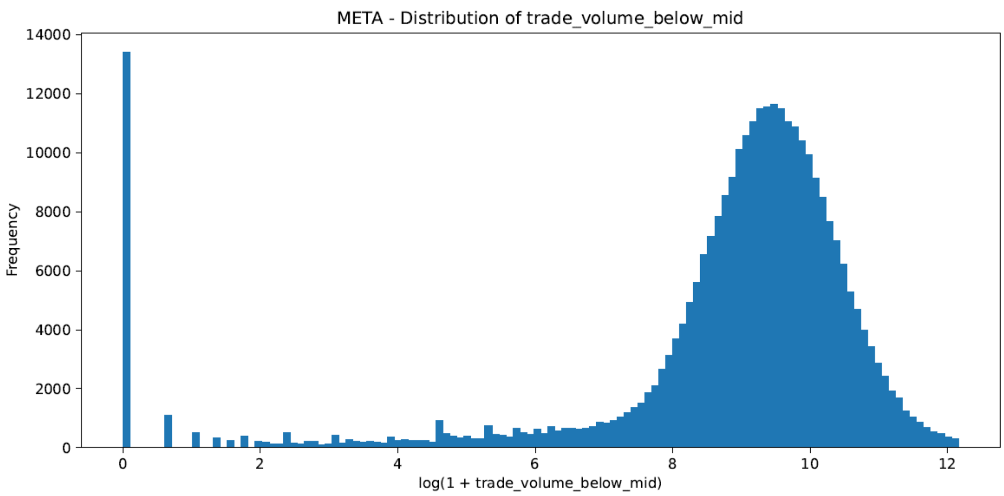

Figure 6: Trading volume below the quoted midpoint.

Figure 5 behaves differently. Its zero mass is stronger, and its positive-volume region is lower and more dispersed than the above-midpoint and below-midpoint distributions. The active midpoint-trading mass is less sharply concentrated and does not reach the same high-volume shape as the two directional side-volume measures.

Statistical drift modeling Let the latent log-price process satisfy

$$dX_t = \mu_t \, dt + \sigma_t \, dW_t,$$

where μ_t is the local drift component and σ_t is instantaneous volatility. The price for diffusion model is log-transferred. The null and alternative hypotheses are same with (10),(11).

Under the null hypothesis, the latent price process is a zero-mean diffusion over the testing window. Under the alternative, the process is a nonzero-mean diffusion, meaning that price changes are due to systematic local drift component. When the null hypothesis is accepted, we conclude the case as noise dominant, likewise when null hypothesis is rejected, we conclude the case as drift dominant.

Figure 7 illustrates the diffusion-based market-state screening pro-

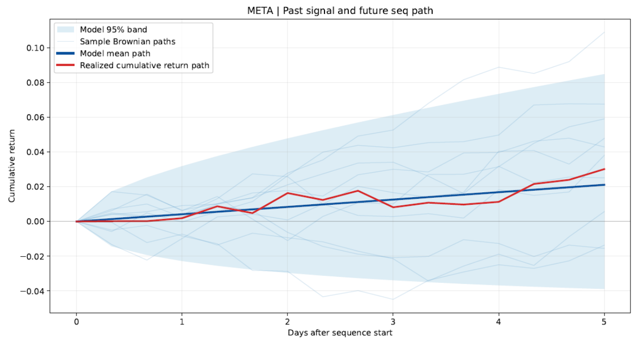

Figure 7: Diffusion-based market-state screening result.

cedure modeling. The light gray paths represent simulated Brownian benchmark paths under a diffusion model, while the blue region reports the model-implied 95% band. The dark blue line gives the model mean path, and the red line shows the realized path.

5.1 Methodology

The empirical design follows a fixed sequence. First, using the constructed high-frequency price series, we fit the diffusion-based framework and apply the hypothesis-testing structure to classify each pre-signal price sequence as drift-dominant or noise-dominant. This step defines the baseline market state. Second, conditional on this baseline state, we examine whether abnormal trading volume appears at any location relative to the bid–ask midpoint: above the midpoint, at the midpoint, or below the midpoint. Third, future return, realized volatility, and price-excursion outcomes are compared across these abnormal-volume cases and the no-abnormal-volume benchmark within the same baseline state. In this design, the main question is whether abnormal within-spread volume follows the outcome pattern implied by the baseline market state, or instead marks a systematic deviation from that baseline.

The central empirical hypothesis is that abnormal trading volume at different locations relative to the bid–ask midpoint can deviate the future price dynamics from before, thus can potentially have predictive power, after conditioning on the baseline market state.

The main outcome variables are measured over the future horizon H. These include:

1. future cumulative return over horizon H;
2. future realized volatility over horizon H;
3. future price-excursion measures, such as maximum favorable and adverse movement over horizon H.

6. Results

The empirical results are organized around two comparisons. The first comparison evaluates whether the realized future paths after abnormal-volume days follow the diffusion-model continuation implied by the pre-signal sequence. The second comparison compares future cumulative returns, upside excursions, downside excursions, and realized volatility across three groups: abnormal volume above the midpoint, abnormal volume below the midpoint, and no abnormal volume.

Throughout this section, "realized avg: abn-above" denotes the ground-truth empirical average future sequence after days when the relevant drift condition is satisfied and abnormal volume is detected only above the midpoint. "Realized avg: abn-below" denotes the corresponding empirical average future sequence after abnormal volume is detected only below the midpoint. "Realized avg: no-abn-vol" denotes the empirical average future sequence after the same drift condition is satisfied but no abnormal volume is detected. These realized-average paths do not impose the diffusion model. The "abn continuation" path is the diffusion-model continuation implied by pre-sequence diffusion parameters estimated from abnormal-volume cases. The "no abn continuation" path is the corresponding diffusion-model continuation implied by no-abnormal-volume cases. The shaded "abn continuation 95%" region is the 95% diffusion-model confidence band around the abnormal-volume continuation path.

Figure 8 reports the average future paths after pre-signal sequences in which drift is not detected.

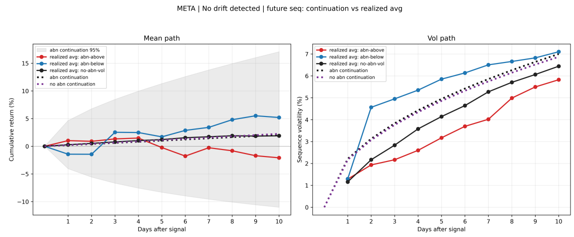

Figure 8: Diffusion baseline average paths for the non-drifting case.

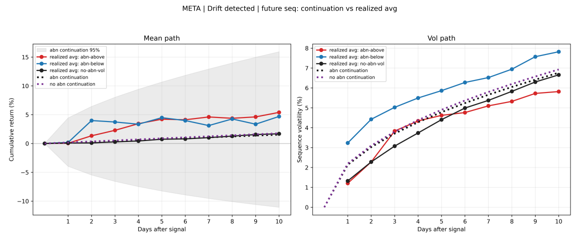

Figure 9: Diffusion baseline average paths for the drifting case.

In the non-drifting case, both the realized future paths and the diffusion-continuation paths are only mildly positive around the baseline. Since drift is not detected before the event, the continuation benchmark is close to a non-directional path rather than a strong trend. The main pattern is that one-sided abnormal volume is followed by a future path that moves more strongly on the opposite side. When abnormal volume appears above the midpoint, the realized future return is weaker than the no-abnormal-volume baseline and turns more negative over the horizon. When abnormal volume appears below the midpoint, the realized future return moves above the no-abnormal-volume baseline. Thus, in the no-drift condition, abnormal above-midpoint volume is followed by weaker subsequent returns, while abnormal below-midpoint volume is followed by stronger subsequent returns.

This pattern suggests that abnormal volume in a non-drifting state pressures subsequent adjustments in the opposite direction. Above-midpoint abnormal volume may reflect short-lived buy-side pressure that is followed by weaker future returns, while below-midpoint abnormal volume may reflect sell-side pressure that is followed by recovery or upward adjustment after the pressure is absorbed.

Figure 9 repeats the comparison for pre-signal sequences in which drift is detected. In this case, both abnormal-volume realized paths are positive, and they remain broadly consistent with the diffusion-continuation paths. Unlike the non-drifting case, the abnormal-volume signal does not mainly appear as an opposite-side adjustment. Instead, abnormal volume tends to move with the detected drift and appears to strengthen the continuation path.

In the drifting case, the realized abnormal-volume paths do not simply follow the model-implied continuation or the realized no-abnormal-volume baseline. Instead, abnormal volume is associated with a positive deviation from these benchmarks. Once drift is detected before the event, both abnormal-above and abnormal-below cases produce realized future paths that move above the no-abnormal-volume realized path and above the diffusion-continuation benchmark. Therefore, abnormal volume is not merely absorbed into the detected drift process; it appears to add further positive movement beyond what the pre-signal diffusion parameters alone would imply.

In the non-drifting case, abnormal volume produces a more side-dependent adjustment: abnormal above-midpoint volume is followed by weaker future returns, while abnormal below-midpoint volume is followed by stronger future returns. In the drifting case, however, the abnormal-volume paths are more positive relative to the baseline. This suggests that abnormal volume amplifies the already directional price environment.

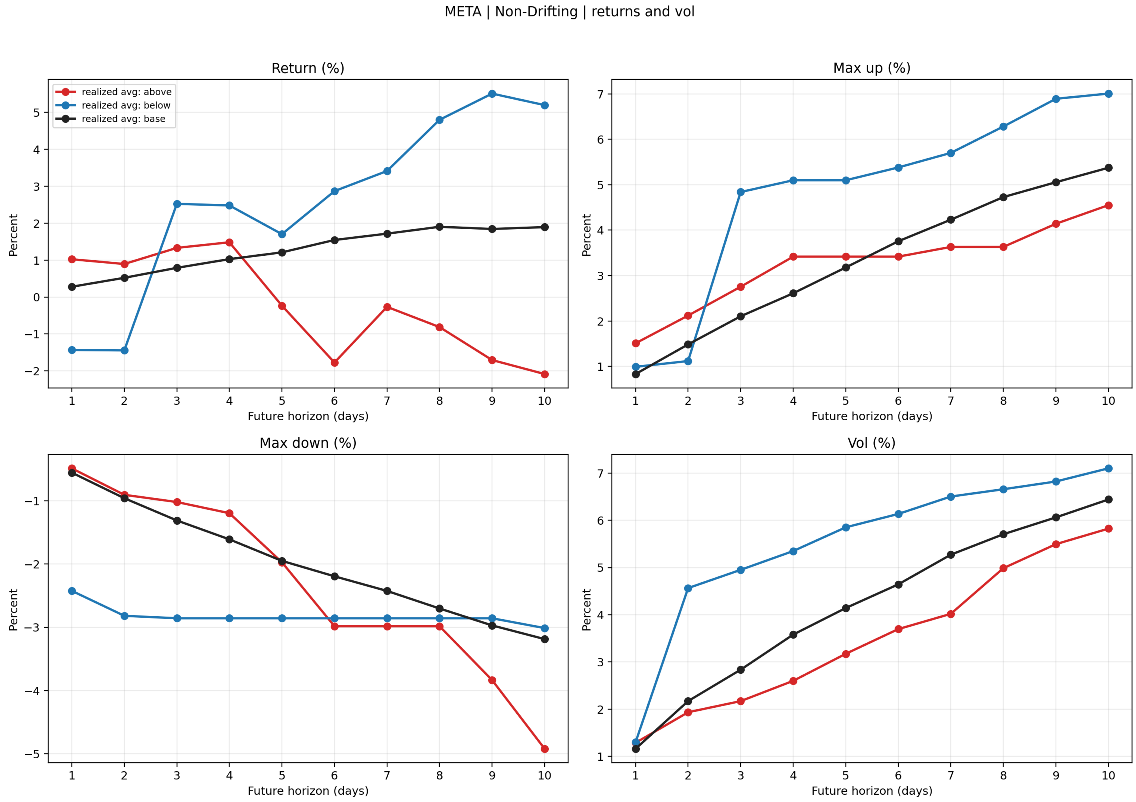

Figure 10: Result for the noise-dominant market condition.

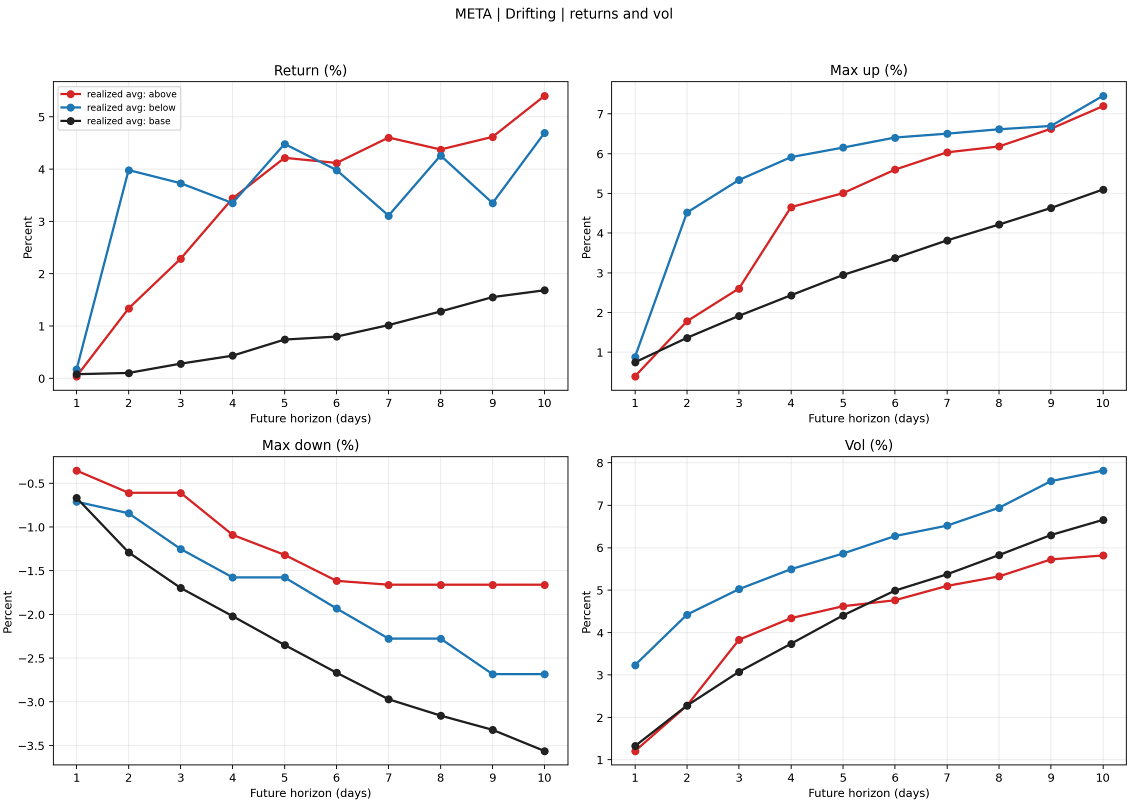

Figure 11: Main empirical result for drift-dominant case.

Figure 10 reports the outcome comparison for the noise-dominant market condition. The benchmark in this figure is the no-abnormal-volume group, which represents pre-signal sequences where no drift is detected and no abnormal within-spread volume appears. Against this benchmark, the abnormal-volume groups show a clear side-specific pattern.

In the cumulative-return panel, the no-abnormal-volume path rises gradually and stays positive. Abnormal below-midpoint volume initially performs worse than this benchmark, but then moves above it and remains higher later in the horizon. Abnormal above-midpoint volume behaves in the opposite way: it does not preserve the early improvement and eventually falls below the benchmark, ending with a negative return. Thus, in noise-dominant states, abnormal below-midpoint volume is followed by stronger later-horizon returns than comparable no-abnormal-volume days, while abnormal above-midpoint volume is followed by weaker later-horizon returns.

The upside-excursion panel reinforces this contrast. The no-abnormal-volume group has a steady increase in favorable movement, but abnormal below-midpoint volume eventually exceeds it. This means the higher return after below-midpoint abnormal volume is not only a small average-return difference; it is also associated with a larger favorable price displacement. Abnormal above-midpoint volume does not show the same pattern. It ends below the no-abnormal-volume benchmark, suggesting that buy-side abnormal volume in a noise-dominant state does not translate into stronger future upside.

The downside-excursion panel adds an important qualification. All three groups experience increasingly negative adverse movement as the horizon expands, but the abnormal above-midpoint group becomes more adverse than the no-abnormal-volume benchmark by the end of the window. Abnormal below-midpoint volume is somewhat more adverse than the benchmark at short horizons, but it does not end as the most negative path. Therefore, the below-midpoint signal combines higher later-horizon return and larger upside without producing the largest final downside excursion. The above-midpoint signal, by contrast, combines weaker return with a more adverse late-horizon downside path.

The realized-volatility panel shows that these return differences are also associated with different path roughness. Abnormal below-midpoint volume has higher realized volatility than the no-abnormal-volume benchmark, while abnormal above-midpoint volume has slightly lower realized volatility. This suggests that the below-midpoint signal is followed by a more active future price path: higher return and larger upside come together with higher realized volatility.

Figure 11 repeats the same comparison for the drift-dominant market condition. Here the benchmark is the no-abnormal-volume group after a drift-detected pre-signal sequence. The main difference from the noise-dominant case is that abnormal volume no longer produces a sharply opposite side-specific pattern. Once drift is detected, both abnormal above-midpoint and abnormal below-midpoint volume are followed by higher cumulative returns than the no-abnormal-volume benchmark.

In the cumulative-return panel, the no-abnormal-volume path rises only moderately. Both abnormal-volume paths move above it for most of the horizon. Abnormal below-midpoint volume separates from the benchmark early, while abnormal above-midpoint volume also remains above the benchmark and finishes higher. This means abnormal volume does not weaken the detected drift. Instead, conditional on a drift-dominant pre-sequence, abnormal within-spread volume is followed by stronger realized continuation than the no-abnormal-volume case.

The upside-excursion panel gives the same interpretation. The no-abnormal-volume benchmark increases gradually, but both abnormal-volume paths produce larger favorable excursions after the initial horizon. Below-midpoint abnormal volume is especially separated from the benchmark in the short and middle horizons, while above-midpoint abnormal volume also stays above the benchmark later. Thus, when drift is already detected, abnormal volume is associated with larger favorable price movement rather than a breakdown of continuation.

The downside-excursion panel shows that this stronger upside is not simply obtained by taking on larger adverse movement. The no-abnormal-volume benchmark becomes more negative than both abnormal-volume groups over the horizon. Both abnormal above-midpoint and abnormal below-midpoint volume have smaller adverse excursions than the benchmark, especially at longer horizons. Therefore, in the drift-dominant state, abnormal-volume cases have higher returns and larger upside than the benchmark while also avoiding the largest downside excursion.

The volatility panel shows that abnormal below-midpoint volume has higher realized volatility than the benchmark. Abnormal above-midpoint volume is closer to the benchmark or near the benchmark at longer horizons. This means the above-midpoint signal is followed by higher return and larger upside without the increase in realized volatility that appears for the below-midpoint signal.

The empirical conclusion is therefore state-dependent and side-dependent. In noise-dominant sequences, abnormal above-midpoint and below-midpoint volume have opposite implications: above-midpoint abnormal volume is followed by weaker later returns, while below-midpoint abnormal volume is followed by stronger later returns, larger upside, and higher volatility. In drift-dominant sequences, abnormal volume on either side is followed by stronger return and larger upside than the no-abnormal-volume benchmark, with below-midpoint volume also carrying higher realized volatility. This supports the central design of the paper: total volume alone is too coarse, because the predictive content appears only after conditioning on both market state and execution location.

7. Implications

The empirical evidence is more naturally interpreted once abnormal volume is treated as a state-dependent liquidity signal rather than as a mechanically bullish or bearish order-flow variable. The results show two main patterns. First, in noise-dominant sequences, abnormal above-midpoint and below-midpoint volume have opposite implications: abnormal above-midpoint volume is followed by weaker later returns, while abnormal below-midpoint volume is followed by higher later returns, larger upside excursions, and higher realized volatility relative to the no-abnormal-volume baseline. Second, in drift-dominant sequences, abnormal volume on either side of the midpoint is followed by higher cumulative returns and larger upside excursions than the no-abnormal-volume benchmark, with below-midpoint abnormal volume also carrying higher realized volatility. These patterns are consistent with the idea that abnormal within-spread volume identifies episodes in which liquidity demand changes the future price path, but the direction of the effect depends on both the baseline market state and the side of the abnormal volume.

The mechanism begins with the economics of immediacy. Liquidity suppliers stand ready to absorb temporary order imbalances created by investors who demand immediate execution [20]. This service is costly because liquidity suppliers take inventory risk, execution risk, and possible adverse-selection risk. They therefore do not absorb abnormal order flow at an unchanged fundamental value; they require compensation for bearing the imbalance. Price pressure arises from this risk-bearing constraint. In this framework, abnormal volume above the midpoint is closer to buy-side immediacy demand, while abnormal volume below the midpoint is closer to sell-side immediacy demand. The midpoint classification is therefore useful because it separates the side of liquidity demand rather than treating all volume as one aggregate object.

In the noise-dominant state, below-midpoint abnormal volume is followed by a future path that rises above the no-abnormal-volume benchmark at later horizons. This suggests that the initial sell-side pressure may create a temporary price concession. Sellers demand immediacy and accept execution closer to the bid; liquidity suppliers or risk-bearing buyers absorb the pressure at a discount; after the pressure is absorbed, the subsequent path can recover and even go beyond the basline. This interpretation is consistent with price-pressure models in which temporary demand shocks move prices because intermediaries require compensation to absorb inventory risk [23]. It is also consistent with fire-sale evidence showing that forced or concentrated selling can create price concessions that later reverse when risk-bearing capital absorbs the imbalance [12].

This interpretation also explains why abnormal below-midpoint volume is followed by higher realized volatility. The below-midpoint signal is not simply a smooth continuation signal. It marks an active liquidity-transfer episode: selling pressure appears, the market absorbs it, and the subsequent path adjusts upward. Such a sequence can generate both larger upside movement and higher realized volatility. Realized volatility measures the roughness of the path, not only its final displacement [5, 2]. Therefore, a path can have stronger positive future returns and higher volatility at the same time if the price moves through an active adjustment process rather than a smooth drift.

The noise-dominant result also clarifies the interpretation of abnormal above-midpoint volume. In a state without detected drift, abnormal above-midpoint volume is followed by weaker later returns relative to the no-abnormal-volume benchmark. Since above-midpoint trades are closer to buy-side immediacy demand, this pattern is consistent with temporary buy-side pressure that does not become durable continuation. Buyers may demand immediate execution near the ask, pushing transaction prices upward in the short run, but if the pre-signal sequence does not contain statistically detected drift, the later path may not support that pressure. The subsequent underperformance relative to the no-abnormal-volume benchmark is therefore consistent with a temporary price-pressure or overreaction channel rather than with informed continuation.

The drift-dominant case has a different interpretation. When drift is already detected before the abnormal-volume event, abnormal volume on either side is followed by higher cumulative returns and larger upside excursions than the no-abnormal-volume benchmark. This means abnormal volume does not weaken the detected drift. Instead, it appears to amplify the already directional environment. The diffusion screen identifies a pre-signal price process with a directional component; abnormal volume then marks the subset of drift-dominant sequences in which trading intensity is unusually elevated. In this state, abnormal volume is better interpreted as reinforcement of the existing price state rather than as a standalone reversal signal.

This result is consistent with models in which trading activity and price dynamics interact through information, liquidity demand, and risk-sharing. If the market is already drifting, abnormal volume can occur because more traders respond to the same directional environment, liquidity demand becomes more concentrated, or intermediaries require larger compensation to absorb order flow in a moving market. Limits to arbitrage make this adjustment imperfect: even if some traders recognize temporary pressure, they may not fully eliminate it because capital constraints and interim loss risk make arbitrage costly [34]. As a result, abnormal volume can accompany stronger continuation in drift states rather than immediately reversing the price path.

Other behavioral mechanisms help explain why the effect differs between noise-dominant and drift-dominant states. In a noise-dominant state, abnormal volume is not supported by a detected directional price process. Above-midpoint abnormal volume can therefore look like temporary buy-side pressure that later fades, while below-midpoint abnormal volume can look like temporary sell-side pressure that later recovers. In a drift-dominant state, however, the market already has a directional component. Extrapolative beliefs and trend-following behavior can make abnormal trading intensity reinforce the existing drift rather than reverse it [3]. Noise-trader risk can also prevent immediate correction, allowing the abnormal-volume episode to remain associated with stronger continuation over the short horizon [13].

The excursion evidence also shows why abnormal volume should not be interpreted only through realized volatility. In the drift-dominant state, the no-abnormal-volume baseline has the larger adverse excursion, while the abnormal-volume groups have higher cumulative returns and larger upside excursions. Therefore, abnormal-volume cases are not simply selecting more dangerous paths. They are selecting paths with a more favorable directional structure. Below-midpoint abnormal volume produces a more active adjustment path, so it carries higher realized volatility. Above-midpoint abnormal volume produces a smoother adjustment path, so its realized volatility is lower even when its return and upside improve. The relevant empirical object is therefore not volatility alone, but the joint behavior of return, upside excursion, downside excursion, and path roughness.

As a result, the paper identifies a conditional volume effect that would be hidden in aggregate volume. Total volume alone cannot distinguish whether abnormal activity occurs above or below the midpoint, and it cannot distinguish whether the event occurs after a noise-dominant or drift-dominant pre-signal sequence. Once these dimensions are separated, the empirical pattern becomes clearer. In noise-dominant states, above-midpoint and below-midpoint abnormal volume have opposite later-return implications. In drift-dominant states, abnormal volume on both sides is associated with stronger continuation than the no-abnormal-volume benchmark. The predictive content of volume therefore depends not only on how much trading occurs, but also on where the trading occurs inside the spread and what price state exists before the abnormal volume appears.

Several limitations should still be noted. The current empirical analysis is based on a limited set of highly traded large-cap stocks, so the results should be interpreted as evidence from liquid equity markets rather than as a universal cross-sectional claim. The abnormal-volume threshold is also based on a rolling quantile rule, which is useful for identifying upper-tail activity but may be sensitive to the lookback window and cutoff level. Future work should test whether the same state-dependent pattern holds across a broader cross-section of assets, different liquidity environments, and alternative abnormal-volume thresholds.

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